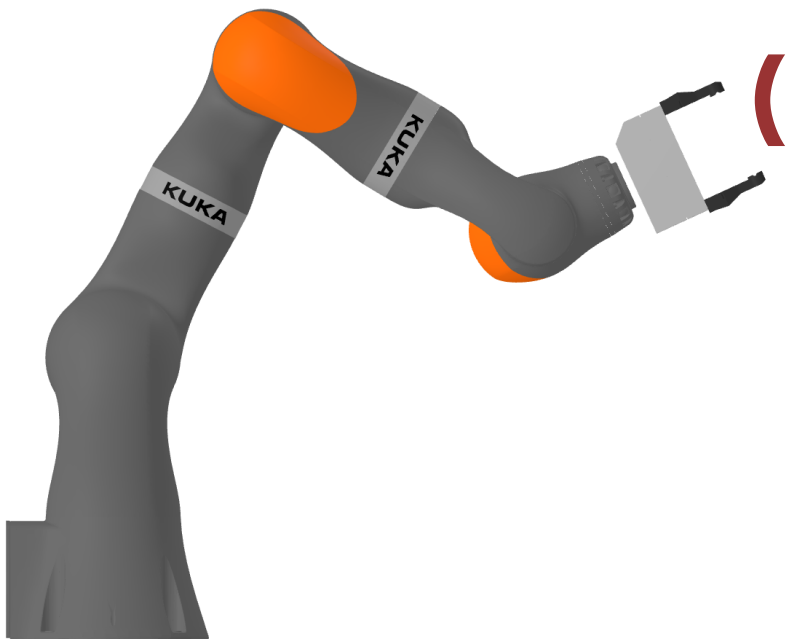


Deep Perception (for manipulation)

Part 2

MIT 6.4210/2

Robotic Manipulation

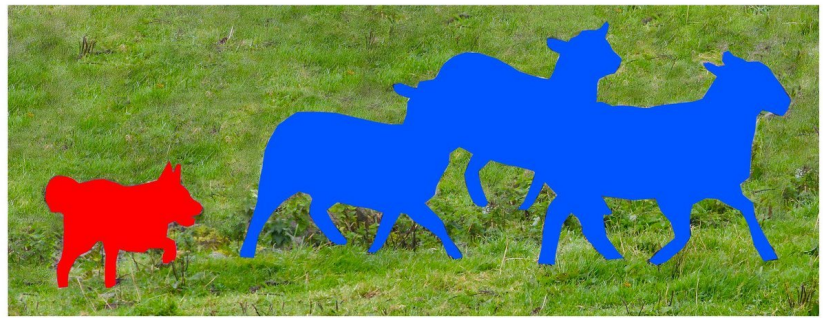
Fall 2025, Lecture 17

Massachusetts Institute of Technology

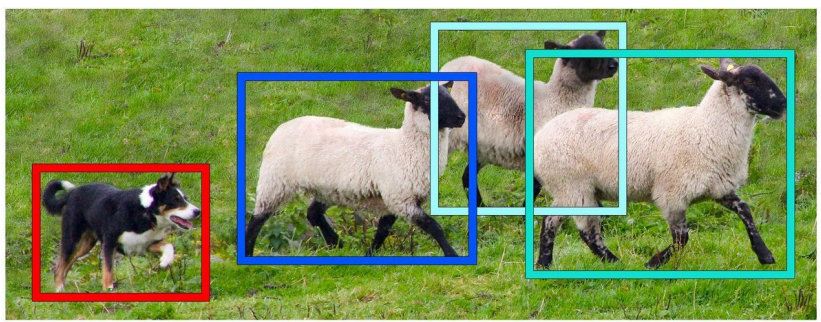
Traditional computer vision tasks



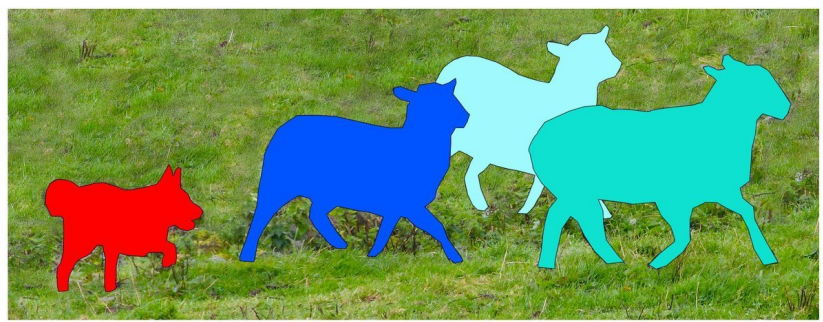
Image Recognition



Semantic Segmentation



Object Detection

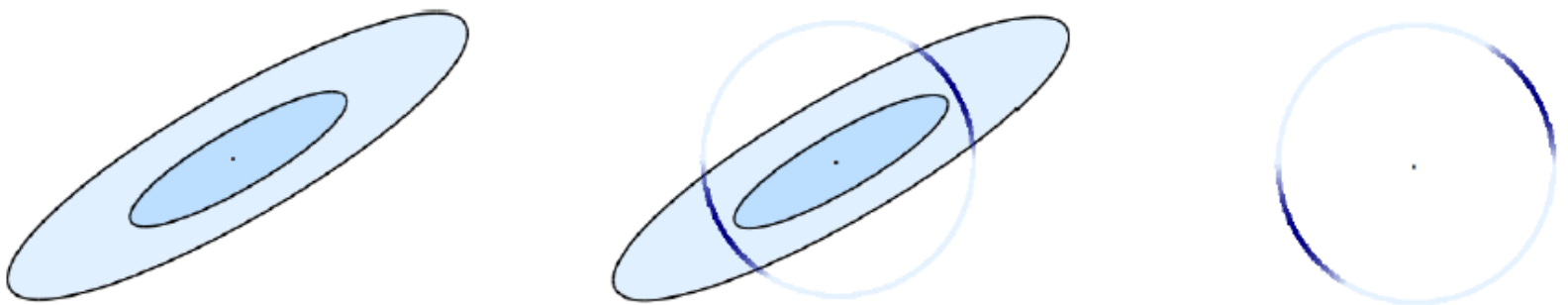


Instance Segmentation

A sample annotated image from the COCO dataset

Bingham distribution (over unit quaternions)

in 2D:

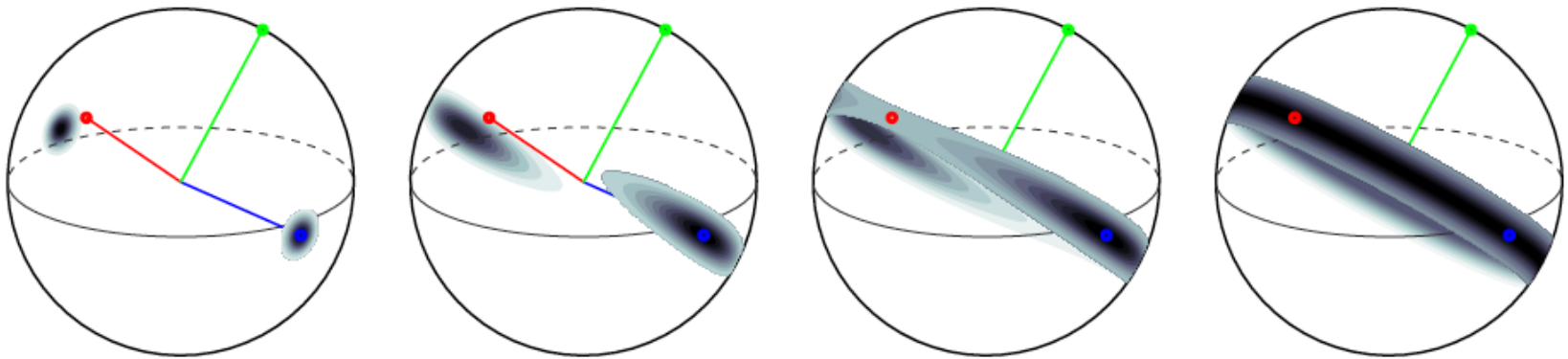


(a) original Gaussian (b) intersection with $\mathbb{S}^1$ (c) resulting Bingham

image from Jared Glover's PhD thesis, 2014

The Quaternion Bingham Distribution, 3D Object Detection, and Dynamic Manipulation

Bingham distribution (over unit quaternions)



(a) $\Lambda = (-100, -100)$ (b) $\Lambda = (-100, -5)$ (c) $\Lambda = (-100, -2)$ (d) $\Lambda = (-100, 0)$

Figure 2-4: The progression of λ_2 towards zero.

from Jared Glover's PhD thesis, 2014

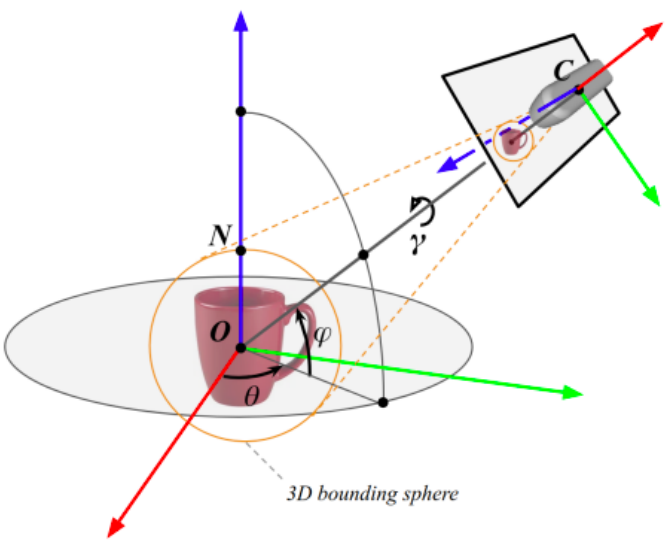


Fig. 2: KOSNet pose representation. The azimuth θ and

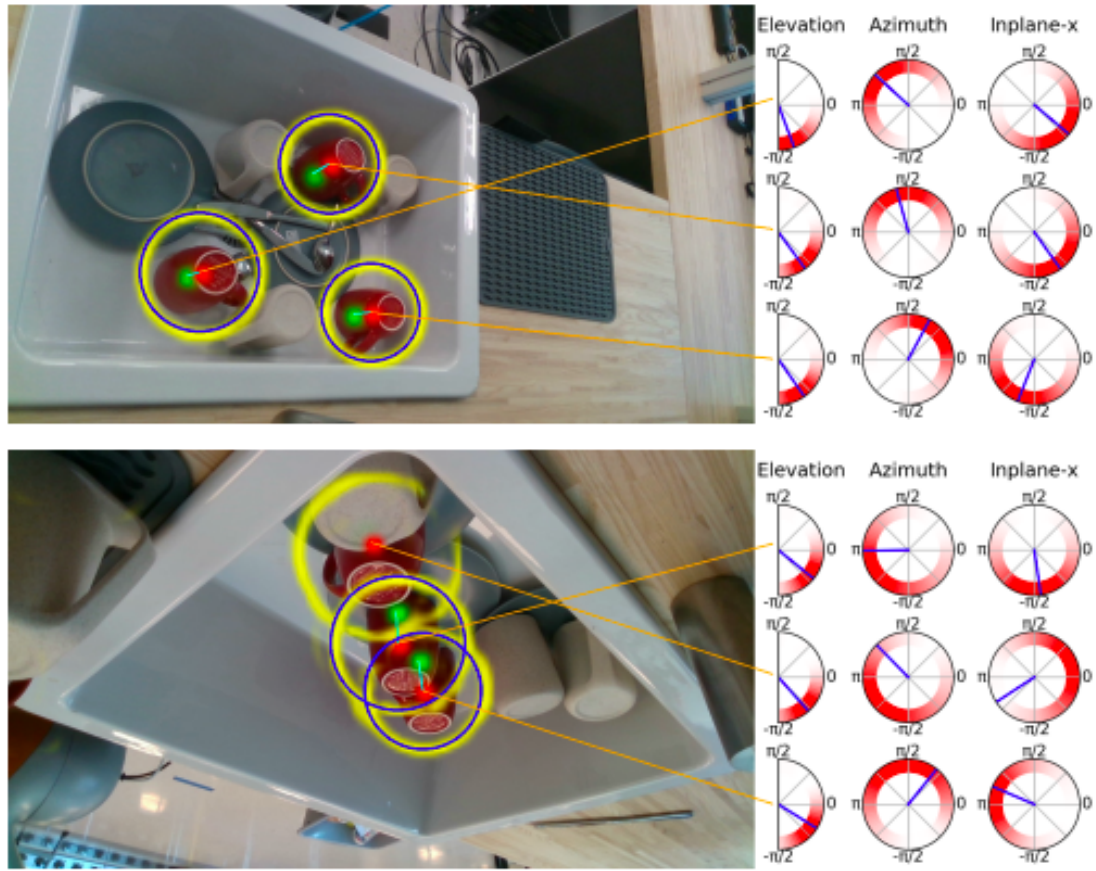
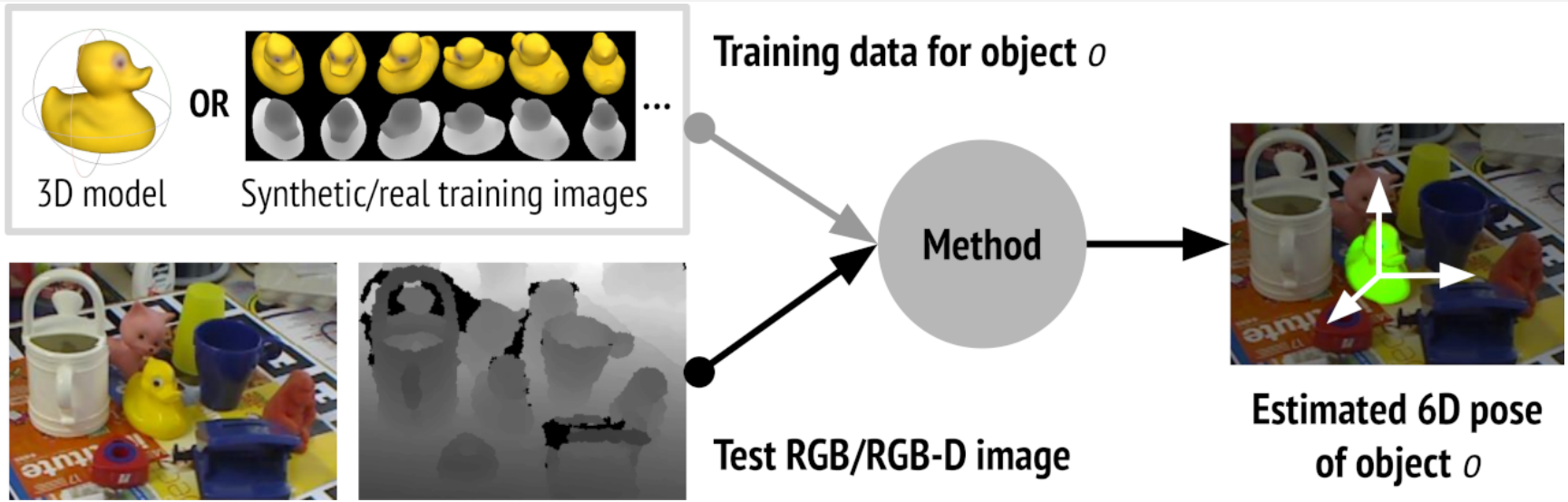


Fig. 5: KOSNet results on TRI Kitchen v1 dataset. Belief

6D Object Pose Estimation Challenge



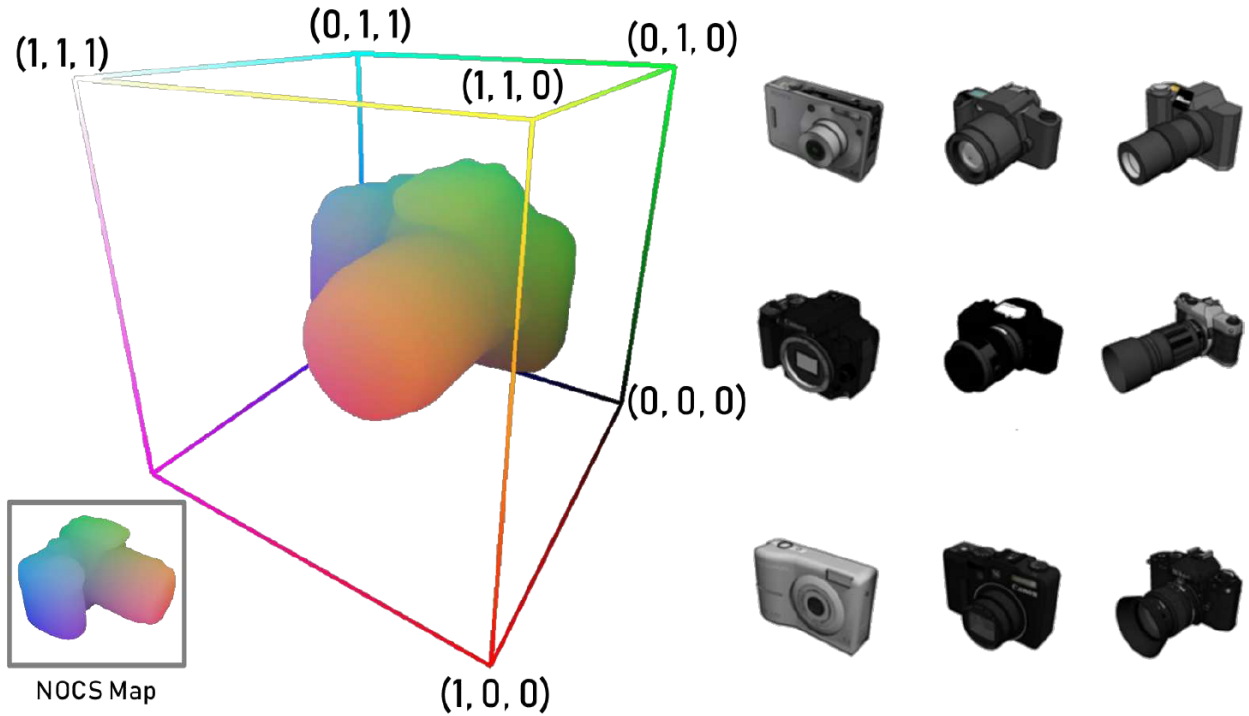
- Until 2019, geometric pose estimation was still winning*.
- In 2020, CosyPose: mask-rcnn + deep pose estimation + geometric pose refinement was best.

* - partly due to low render quality?

https://docs.google.com/presentation/d/1ByJh1Ok2VF5QiY2DsNGMNGjvfLGVUzIVMS9edHR77rQ/edit?slide=id.g3048515829d_1_1527#slide=id.g3048515829d_1_1527

Category-level Pose Estimation

"NOCS"



H. Wang, S. Sridhar, J. Huang, J. Valentin, S. Song, and L. J. Guibas, “Normalized Object Coordinate Space for Category-Level 6D Object Pose and Size Estimation,” in *2019 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, Long Beach, CA, USA, Jun. 2019, pp. 2637–2646, doi: 10.1109/CVPR.2019.00275.

Grasp Pose Detection in Point Clouds

The International Journal of
Robotics Research
2017, Vol. 36(13–14) 1455–1473
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sagepub.co.uk/journalsPermissions.nav
DOI: 10.1177/0278364917735594
journals.sagepub.com/home/ijr
SAGE

Andreas ten Pas[1], Marcus Gualtieri[1], Kate Saenko[2] and Robert Platt[1]

Dex-Net 2.0: Deep Learning to Plan Robust Grasps with Synthetic Point Clouds and Analytic Grasp Metrics

Jeffrey Mahler*, Jacky Liang*, Sherdil Niyaz*, Michael Laskey*, Richard Doan*, Xinyu Liu*,
Juan Aparicio Ojea†, and Ken Goldberg*
*Dept. of EECS, University of California, Berkeley

2018 IEEE International Conference on Robotics and Automation (ICRA)
May 21-25, 2018, Brisbane, Australia

Robotic Pick-and-Place of Novel Objects in Clutter with Multi-Affordance Grasping and Cross-Domain Image Matching

Andy Zeng[1], Shuran Song[1], Kuan-Ting Yu[2], Elliott Donlon[2], Francois R. Hogan[2], Maria Bauza[2], Daolin Ma[2],
Orion Taylor[2], Melody Liu[2], Eudald Romo[2], Nima Fazeli[2], Ferran Alet[2], Nikhil Chavan Dafle[2], Rachel Holladay[2],
Isabella Morona[2], Prem Qu Nair[1], Druck Green[2], Ian Taylor[2], Weber Liu[1], Thomas Funkhouser[1], Alberto Rodriguez[2]
[1]Princeton University [2]Massachusetts Institute of Technology

https://www.youtube.com/embed/kfe5bNt35ZI?mute=1&enablejsapi=1&rel=0

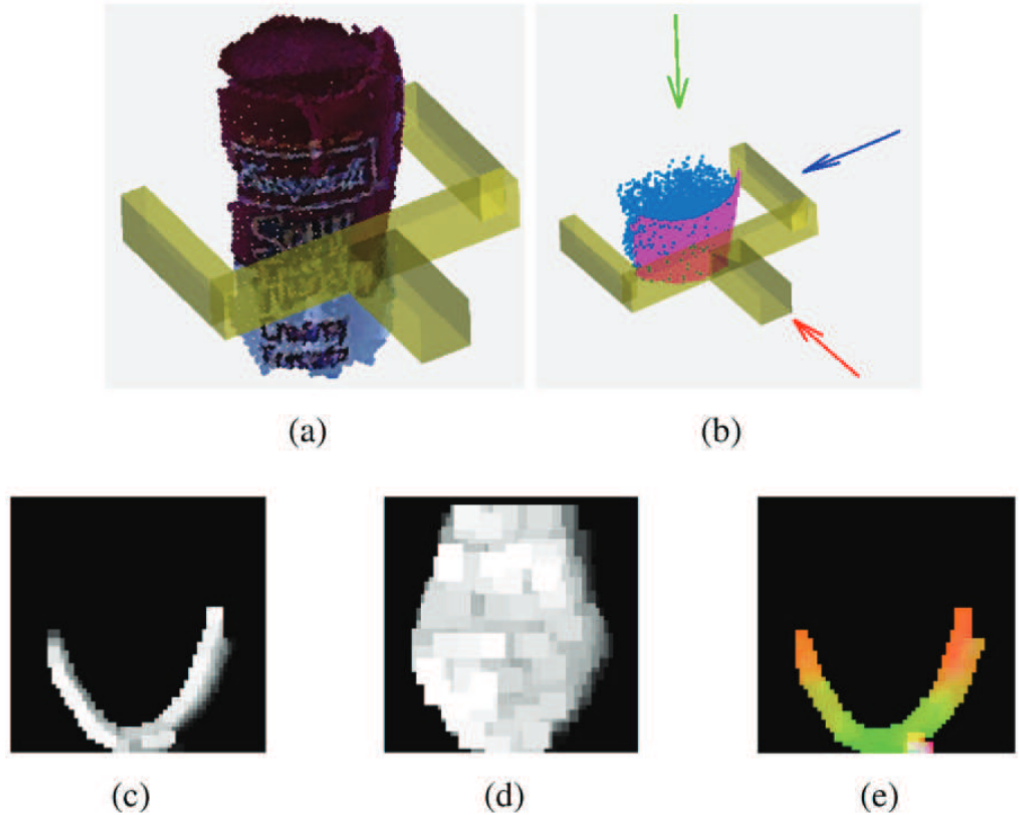


Fig. 5. Grasp representation. (a) A grasp candidate generated from partial point cloud data. (b) Local voxel grid frame. (c)–(e) Examples of grasp images used as input to the classifier.

https://www.youtube.com/embed/i6K3GI2_EgU?start=25&enablejsapi=1

https://dexgraspanything.github.io/


All manipulated objects
in this video have
not been seen before
1x speed

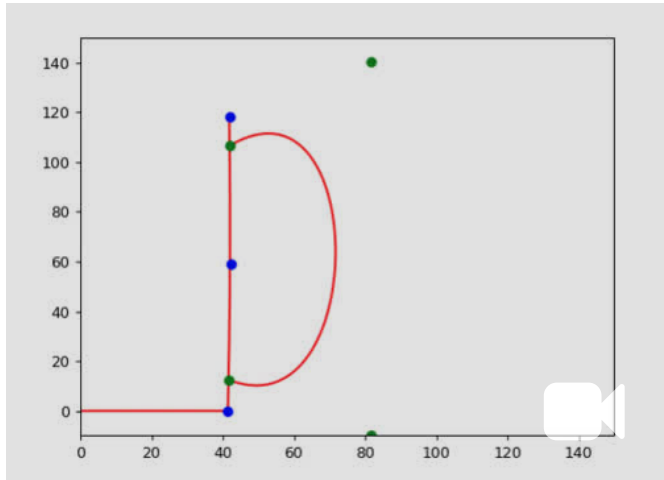
"Procedural dishes"!



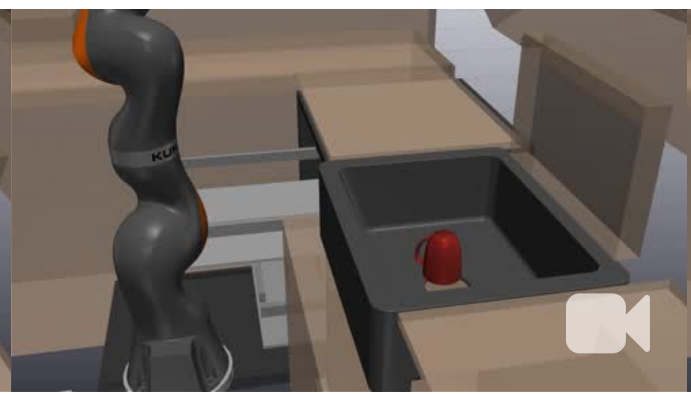






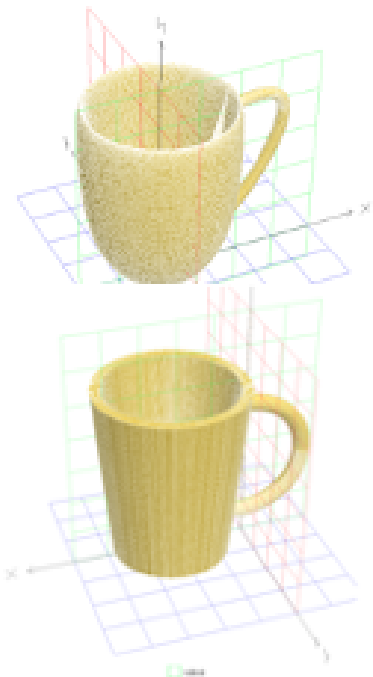








"Procedural dishes"



+

TOYOTA
RESEARCH INSTITUTE



→



SE(3) pose is difficult to generalize across a category



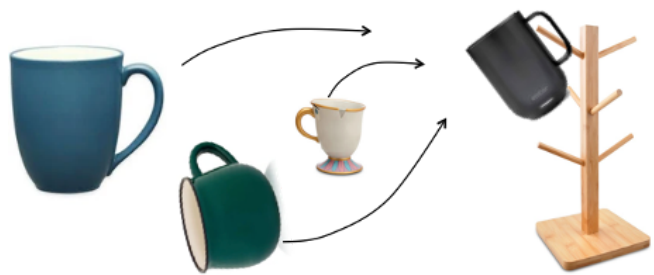




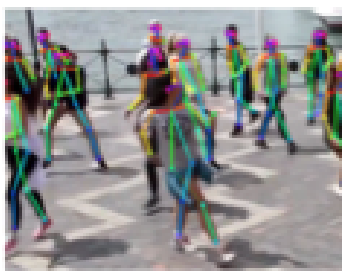
So how do we even specify the task?

What's the cost function?

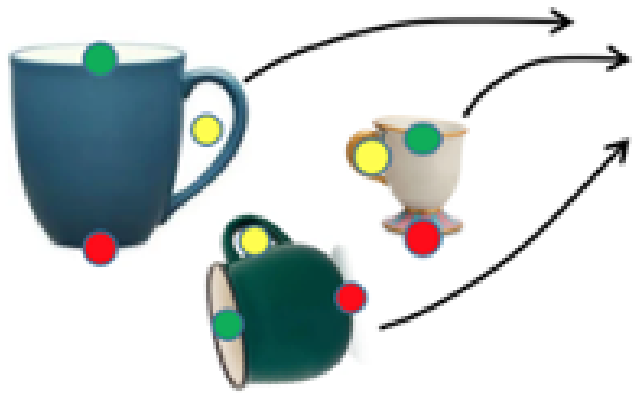
(Images of mugs on the rack?)



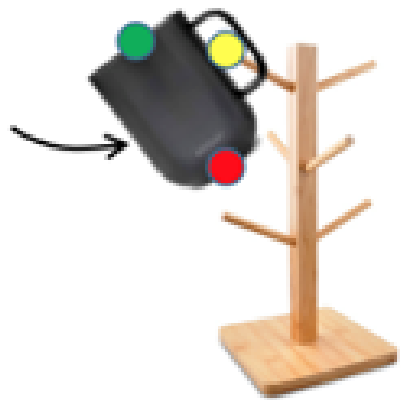
3D Keypoints provide rich, class-general semantics



OpenPose



Constraints & Cost on Keypoints

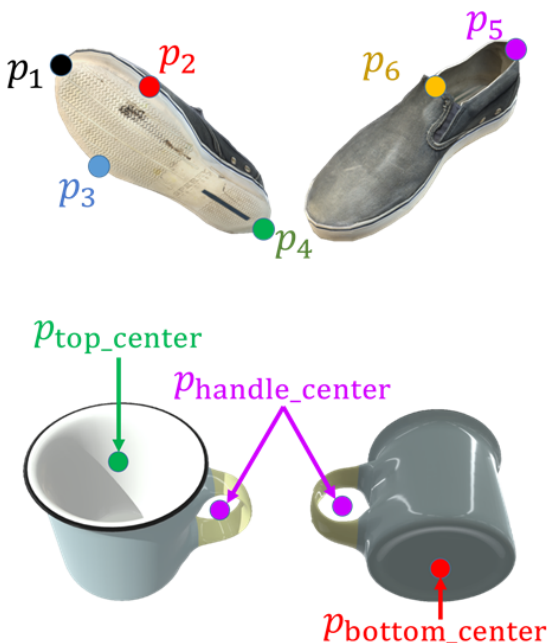


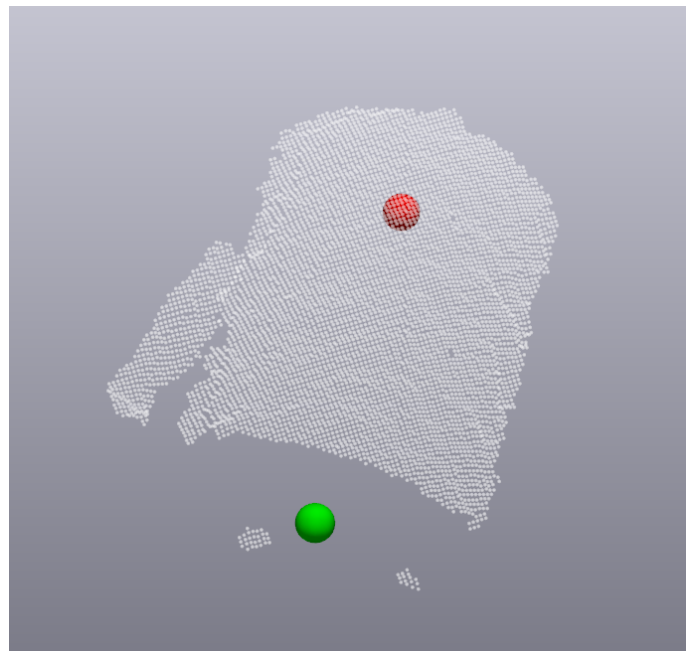
… and robust performance in practice



Lucas Manuelli*, Wei Gao*, Peter R. Florence and Russ Tedrake. kPAM: KeyPoint Affordances for Category Level Manipulation. ISRR 2019

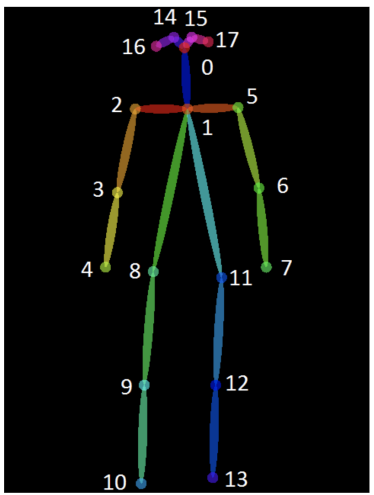
Keypoints are not a sufficient representation





Keypoint "semantics" + dense 3D geometry


1x speed

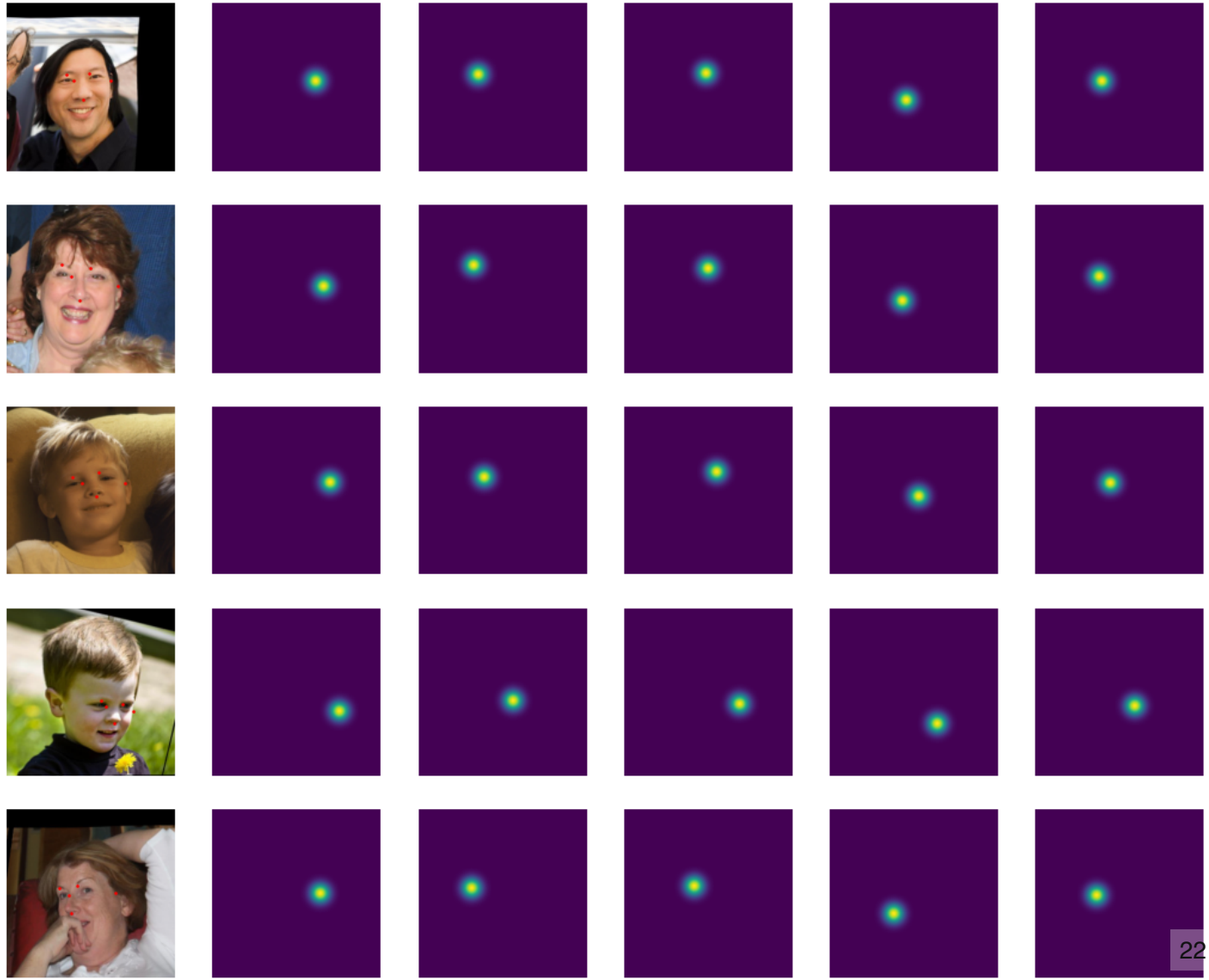




https://nanonets.com/blog/human-pose-estimation-2d-guide/

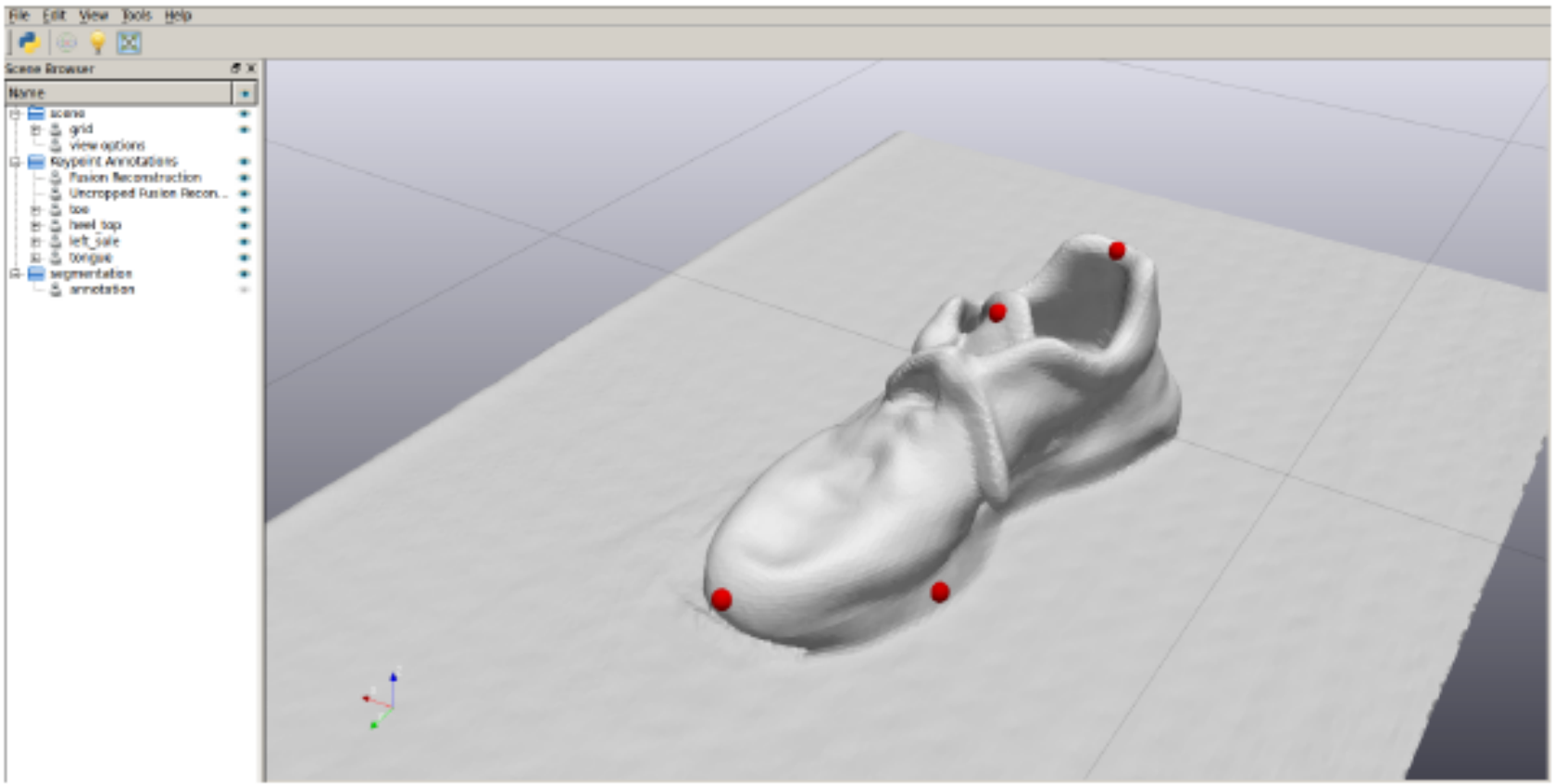
https://keypointnet.github.io/

Typically don't predict keypoints directly; predict a "*heatmap*" instead



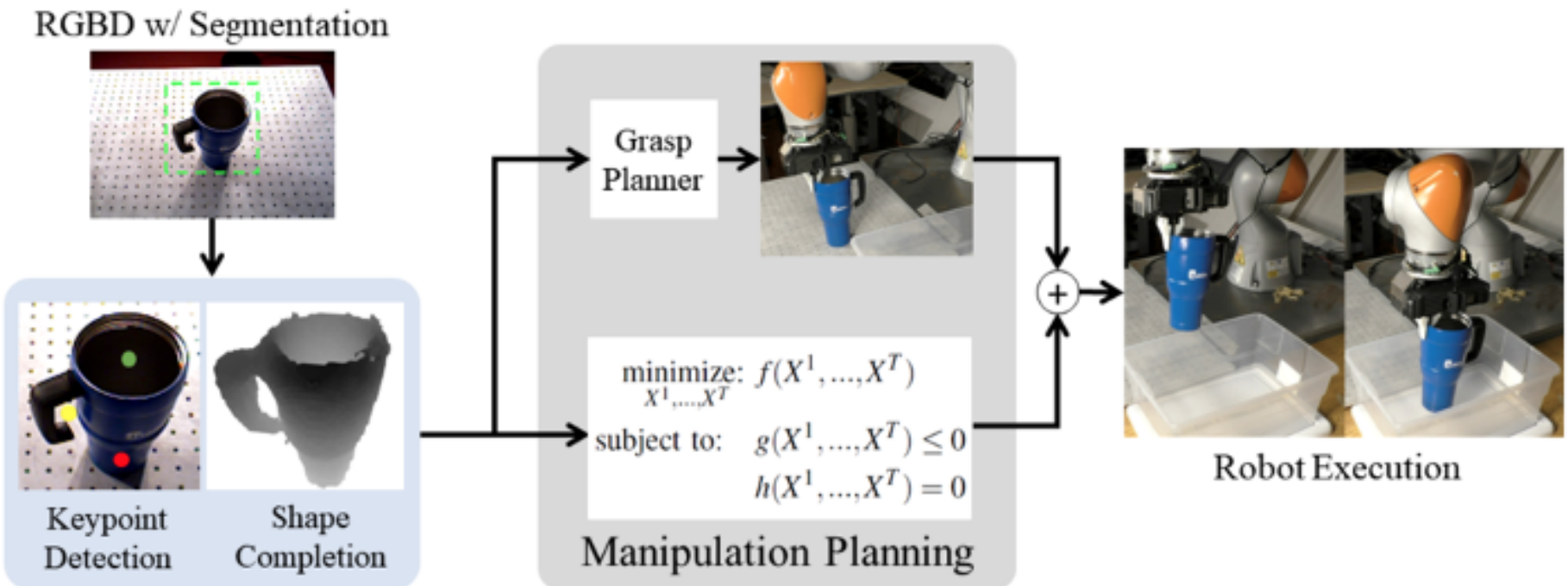


Dense reconstruction + annotation GUI

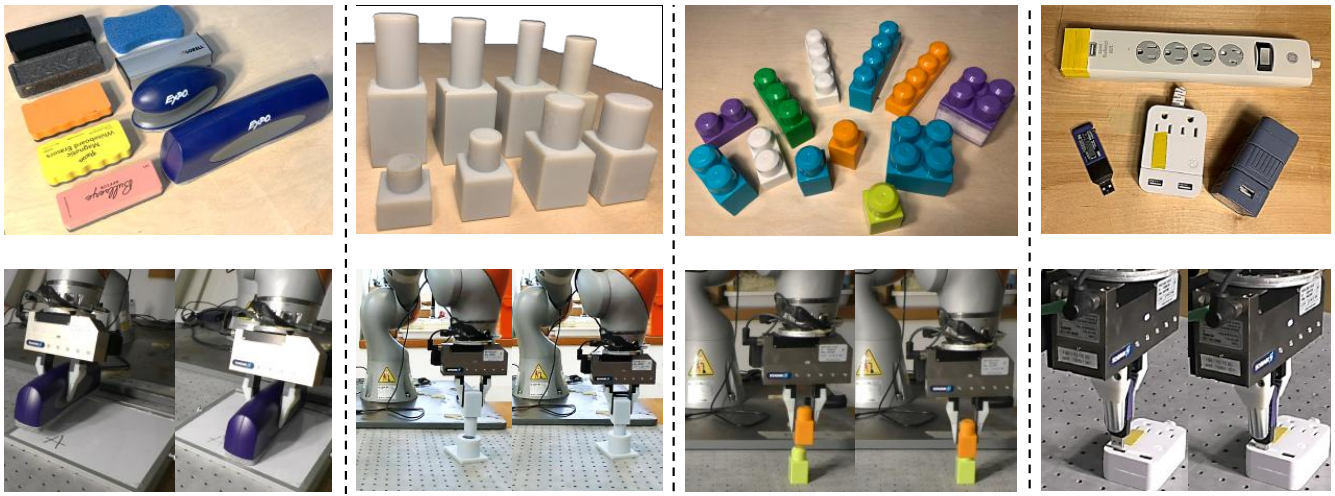


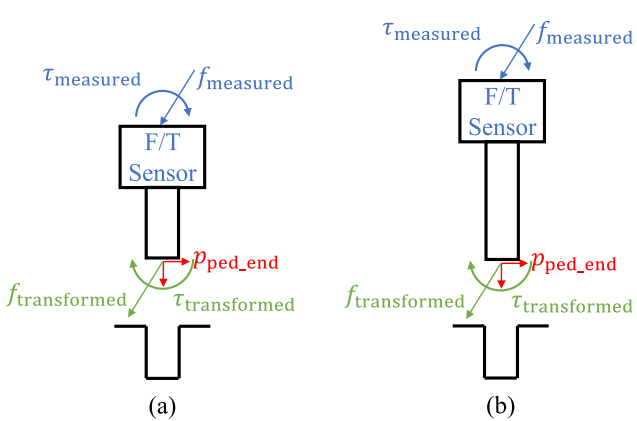
+ shape completion network (kPAM-SC)

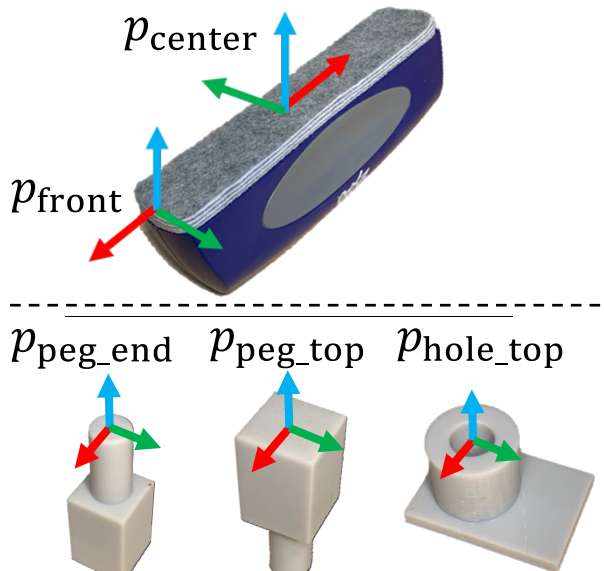
to include collision-avoidance constraints



+ force control?







Feedback Control for Category-Level Robotic Manipulation

Wei Gao, Russ Tedrake
CSAIL, Massachusetts Institute of Technology

So far, keypoints are *geometric* and ***semantic***
(mug handle, front toe of shoe), but required human labels

If we forgo semantics, can we **self-supervise**?

https://kalm-il.github.io/

https://f3rm.github.io/

https://www.youtube.com/embed/PA9rWWVWsc4?mute=1?enablejsapi=1

Retrographic sensing for the measurement of surface texture and shape

Micah K. Johnson
MIT
kimo@mit.edu

Edward H. Adelson
MIT
adelson@csail.mit.edu

Abstract

photometric stereo algorithm tailored to the sensor, we reconstruct the surface, which is rendered in Fig. 1(c).







https://www.youtube.com/embed/aKoKVA4Vcu0?start=5&mute=1&rel=0&enablejsapi=1

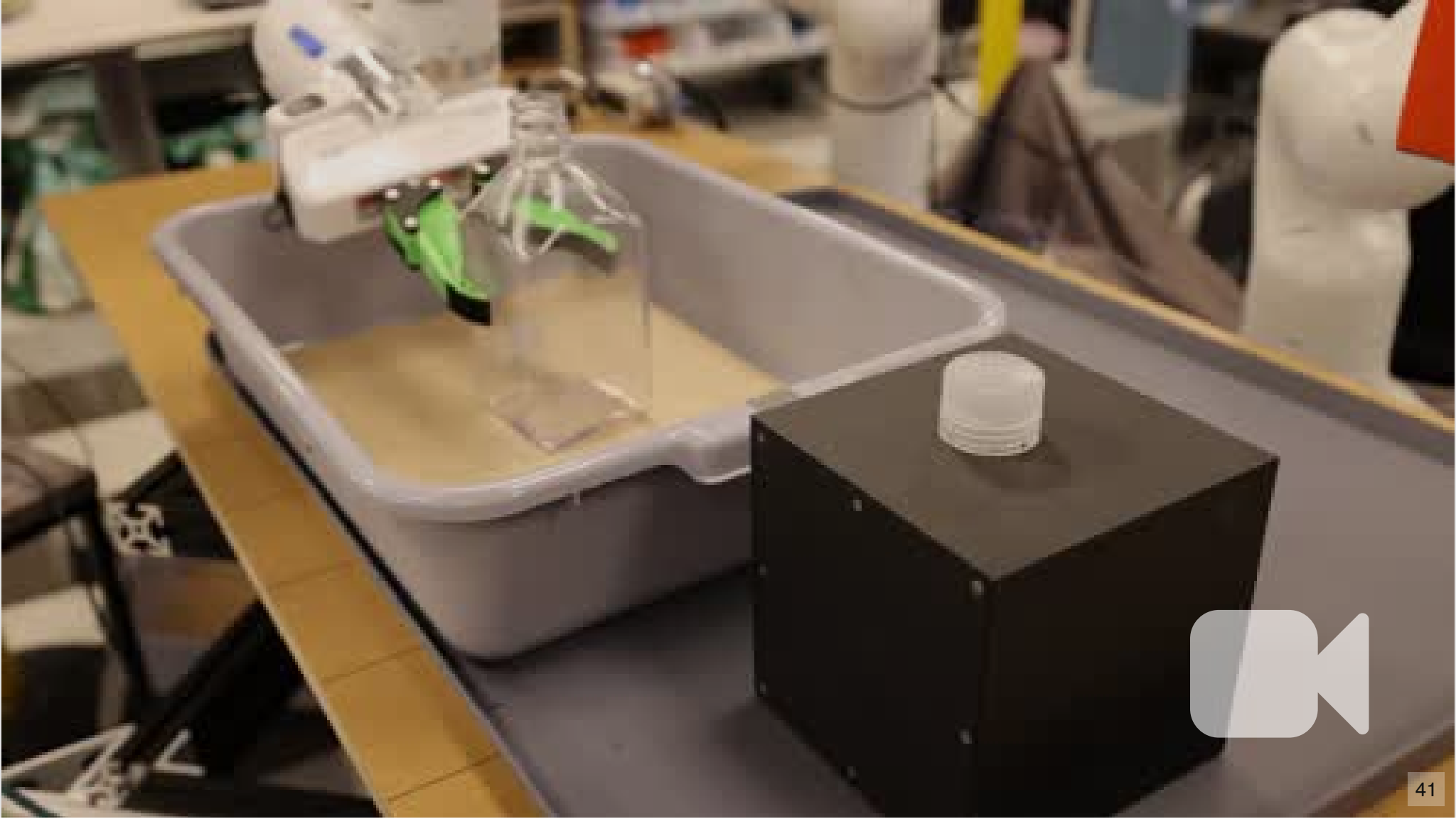
https://www.youtube.com/embed/BIW_jq3dOEE?enablejsapi=1

https://www.youtube.com/embed/f59FoS-hV7c?mute=1&rel=0&enablejsapi=1



https://www.youtube.com/embed/2-YuIfspDrk?start=152&mute=1&rel=0&enablejsapi=1



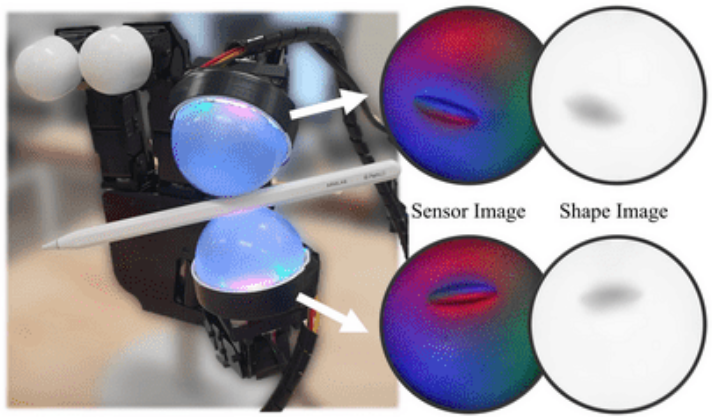


DenseTact: Optical Tactile Sensor for Dense Shape Reconstruction

Won Kyung Do, Monroe Kennedy
Mechanical Engineering, Stanford University

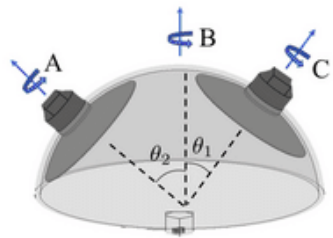
Assistive Robotics and Manipulation Laboratory

Introduction - DenseTact



1. A new 3D-shaped, vision-based soft tactile sensor design with a very soft elastomer contact surface with low construction cost and high sensing resolution
2. Novel calibration process for model which performs 3D shape reconstruction from single image with high resolution in real time, and
3. Adapted deep neural network for estimating sensor surface from interior sensor image.

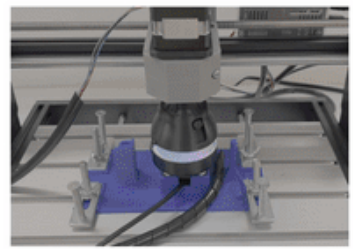
Data Generation



Hemispherical indicator



Different type of indenters



Generated depth data by contacting a known 3D surface model (0.1mm error, about 30,000 configurations).

https://www.youtube.com/embed/yI6WDzfYD8Q?enablejsapi=1&rel=0&start=130&mute=1

https://pages.gelsight.com/digit-tactile-sensor